LAB 18

# Diagnostics Stage: Automated Bias Checks before Go-Live

| Item | Detail |
|---|---|
| Topic | Topic 4: Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis |
| Pipeline stage | Diagnostics |
| TSC reference | A9, K8 |
| Duration | 20 min |
| Simulator page | http://localhost:8000/#/research/diagnostics?strategy=custom_lab18_biased_smoothed_momentum |
| Mode | No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only. |

*Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.*

## Scenario

A junior quant sends the investment committee a 'smoothed momentum' backtest on SPY with a Sharpe ratio above 4 and asks to go live. The desk responds by adding a diagnostics stage to the pipeline: every strategy - including its own ema_cross - must pass automated checks for look-ahead, cost drag, overfitting and survivorship before it may trade, even on paper.

## Objective

Run the automated diagnostics on a deliberately biased strategy, fix it, and re-run.

## Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 18. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

## Lab Steps - In the Trading Simulator

*No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 18: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.*

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### Step 1. Diagnose the biased exhibit

Research > Diagnostics on 'Lab 18 exhibit: biased smoothed momentum' (seeded custom strategy).

Open: http://localhost:8000/#/research/diagnostics?strategy=custom_lab18_biased_smoothed_momentum (Research > Diagnostics (Lab 18 exhibit))

*Check: Look-ahead (features) FAIL: rolling(center=True) uses future bars. Verdict BLOCK.*

### Step 2. Fix it in the Strategy Builder

Open the exhibit in the Strategy Builder and change the single setting center=True to center=False (or let AI fix it: use the Lab 18 Strategy Builder prompt in the Prompt Library). Validate, Save.

Open: http://localhost:8000/#/builder?key=custom_lab18_biased_smoothed_momentum (Strategy Builder (exhibit))

*Check: The validation's look-ahead check now passes.*

### Step 3. Re-run diagnostics

Diagnose the fixed strategy; compare the same-bar vs next-bar Sharpe and the cost-drag row.

Open: http://localhost:8000/#/research/diagnostics?strategy=custom_lab18_biased_smoothed_momentum

*Check: No FAIL on look-ahead; cost drag and overfitting WARN explained.*

**Expected in the app: A FAIL on the biased exhibit, a PASS / WARN table for the fixed version and a verdict.**

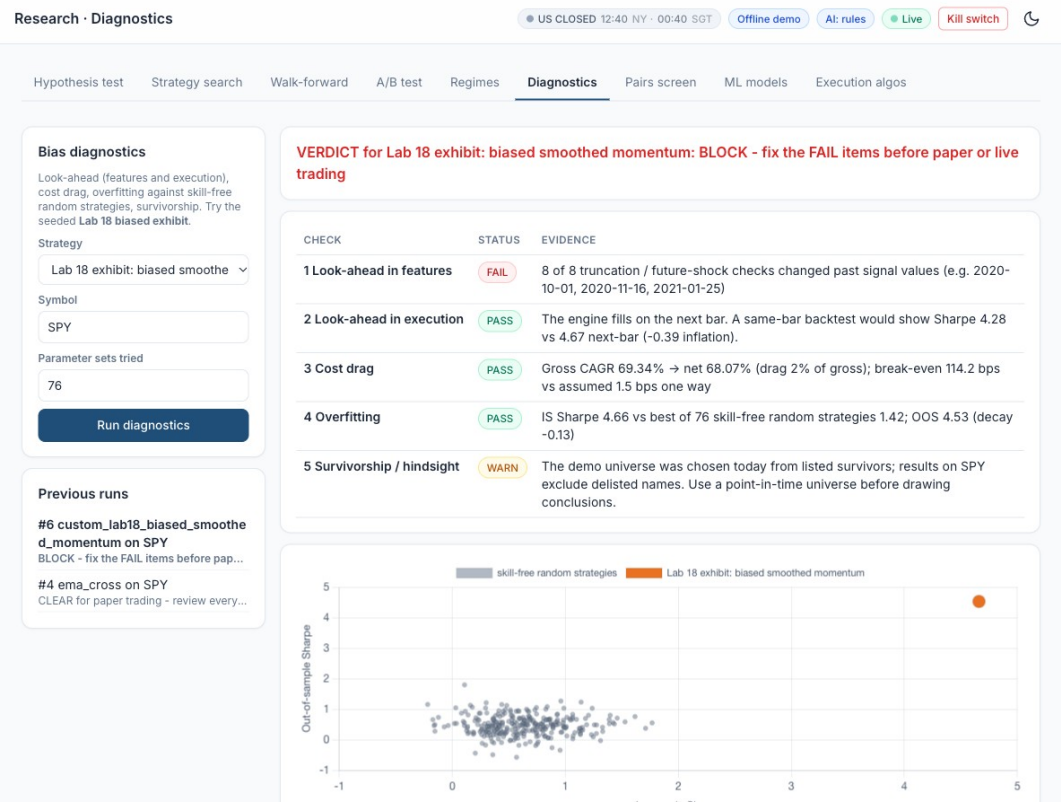



*Figure: Lab 18 in the Trading Simulator.*

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## Optional: Let AI Write the Code

*This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs_lowcode/Prompt_Library.html and .pdf.*

| Diagnose a too-good-to-be-true backtest - run it in Google Colab |
| --- |
| I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: recreate this biased SPY strategy on 2018-2025: signal = 1 when the 5-day rolling mean computed with center=True is rising; strategy return = signal x the same day's return; no costs. Print its CAGR and Sharpe, then fix it one problem at a time - a trailing window, next-day execution, a 0.05% cost - printing CAGR and Sharpe after each fix next to buy-and-hold, and explain each bias in plain English. |

**Open in Claude** | **Open in ChatGPT** | **Open Google Colab**

| Fix the Lab 18 exhibit in the Strategy Builder - paste it into the Strategy Builder |
| --- |
| Here is a strategy from my Trading Simulator's Strategy Builder that looks too good to be true. Find the look-ahead bias, explain it in plain English for a beginner, and give me the corrected code with the same class name and params, changing as little as possible. Use only data up to each bar: the engine trades on the next bar. Here is the code: [paste the exhibit's code from the Strategy Builder here] |

**Open in Claude** | **Open in ChatGPT**

## Debrief

Most 'too good to be true' results come from a short list of causes (K8): look-ahead in features or execution, overfitting, survivorship and ignored costs. Automating the checks and using the verdict as a gate (A9) is what stops an unintended outcome from reaching even a paper account - and the same checks run on the desk's own strategy, not only on other people's.

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